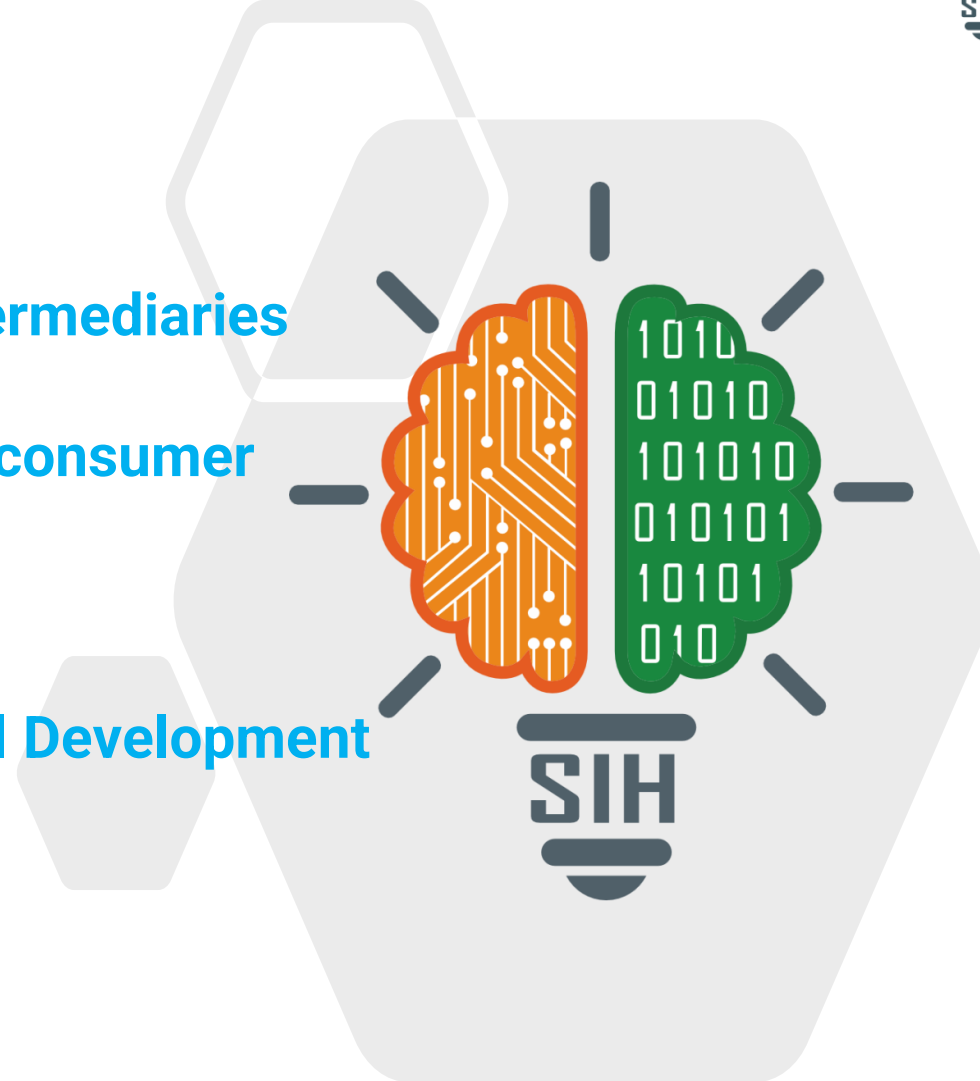


SMART INDIA HACKATHON 2026



- Problem Statement ID - [SIH26033](#)
- Problem Statement Title - [Multiple intermediaries reduce farmers earnings and increase consumer prices.](#)
- Theme - [Agriculture, FoodTech & Rural Development](#)
- PS Category - [Software](#)
- Team ID - [152023](#)
- Team Name (Registered on portal) - [Parallax](#)





The Problem



Farmer receive
33-37%
of consumer price



Consumer pay
2 – 3x
the farm gate price



40% of perishable
product wasted before
reaching market



Multiple intermediaries:
Higher cost, less transparency,
more waste.



It addresses the Problem by



Direct market access
Connects farmers/FPOs directly with consumers and buyers.



Better price discovery
Helps farmers access demand and market information.



Reduce intermediary dependence
Fewer middlemen can improve transparency and farmer earnings.



Lower post-harvest losses
Coordinate logistics helps move produce efficiently and reduce wastage.



Traditional Chain (The Problem)



Farmer



Arhtiyas



Wholesaler



Retailer



Consumer



KrishiSetu Solution



Farmer /
FPO



KrishiSetu



Consumer /
Bulkbuyer



Impact



Higher earnings for farmers.



Fair prices for consumers.



Reduced waste & efficient supply chain.



Stronger rural economy & livelihoods.

Technical Approach

"Technology that connects every step"

Tech Stack

Frontend

HTML CSS JS Bootstrap

Backend

Python Flask REST API

Database

PostgreSQL SQLAlchemy

AI / ML

Pandas NumPy Scikit-learn

Maps & Routing

Leaflet OpenStreetMap OSRM

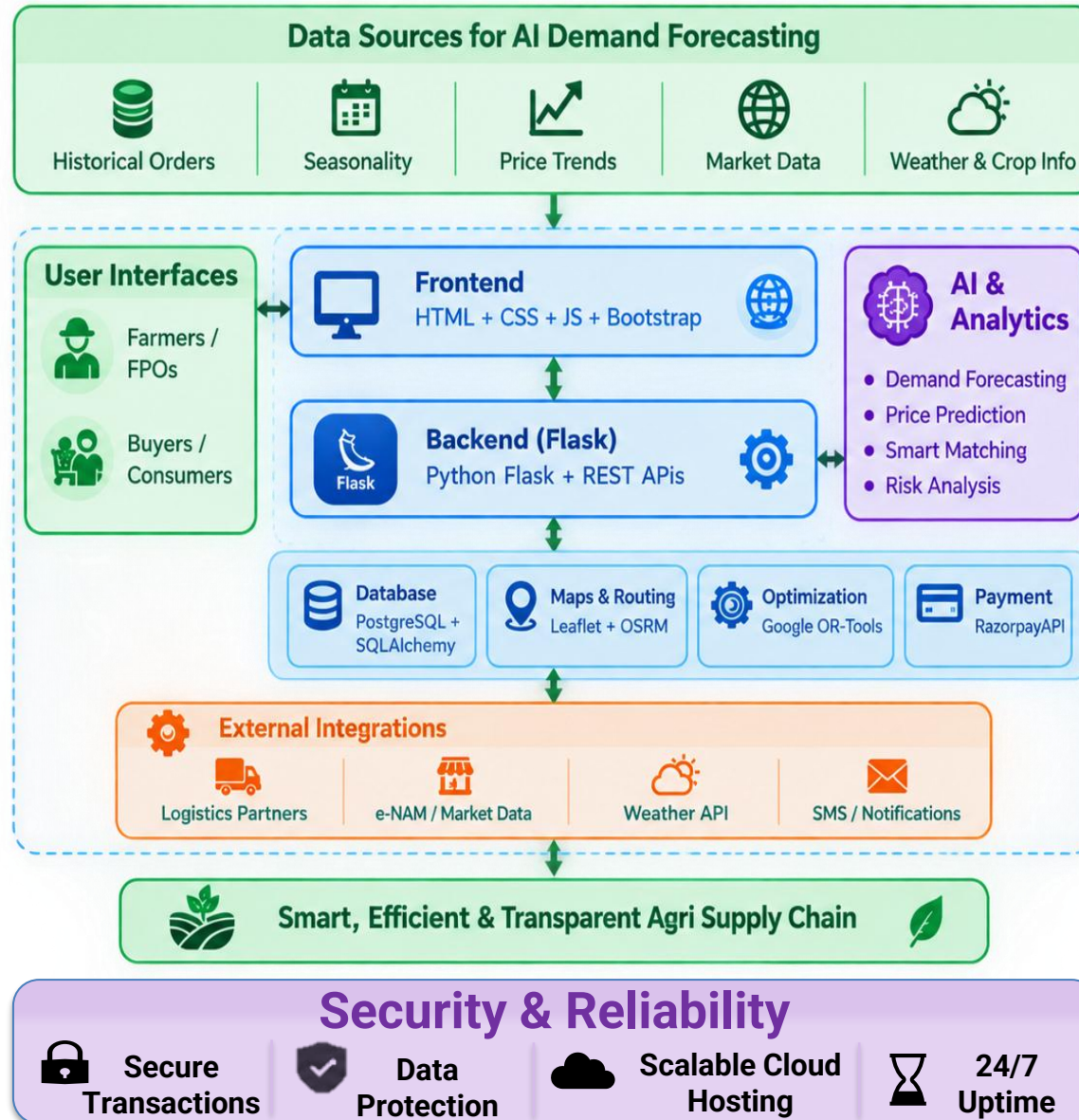
Optimization

Google OR-Tools

Payment

Razorpay

✓ No new hardware required.



System Flow

- 1 Farmers list their produce.
- 2 AI forecasts demand.
- 3 Smart matching with buyers.
- 4 Buyers payment held in escrow.
- 5 Logistics pickup and delivery.
- 6 Quality confirmed → payment released.

Key Features

- ✓ Real time market access & live pricing.
- ✓ AI-based demand forecasting.
- ✓ Smart matching of farmers & buyers.
- ✓ Route optimization for efficient logistics.
- ✓ Secure digital platforms.

Prototype

Prototype video



FINANCIALS



Commission
Per-transaction
fee



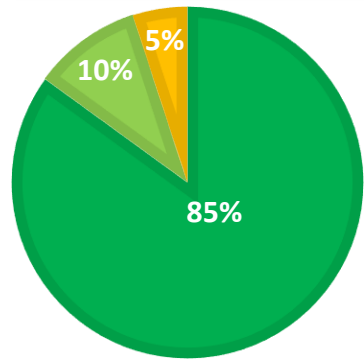
Logistics Fee
Coordination
charge



FPO Tools
Premium
subscriptions

Illustrative Revenue model

Revenue per ₹1,000 order



● **Farmer**
₹850 (85%)

● **Logistics**
₹100 (10%)

● **KrishiSetu
Platform Fee**
₹50 (5%)

Key Takeaway

Farmers receive the majority of the order value, while a small platform fee enables Krishi Setu to continue serving them sustainably.



FEASIBILITY



Existing stack: HTML • CSS • JavaScript • Node.js • Python • PostgreSQL • OR-Tools



Maps + OR-Tools: distance, capacity, pickup/delivery, time windows, routing



86.2% of holdings are small & marginal (≤ 2 ha).



398.35M rural internet subscribers; 44.16 per 100 population (FY 2023–24)



e-NAM currently links 1,522 markets across 23 states + 4 UTs – evidence of a large digital agricultural market ecosystem.

Buildable with existing technology + real market need + practical partner-based operations.



VIABILITY



Revenue Engines: Logistics Fee | BulkBuyer / FPO Tools | Analytics



Public-private partnerships with FPOs & logistics firms enable **asset-light scaling**.



B2B repeat demand → recurring transactions.



Fragmented supply + recurring bulk demand creates a scalable logistics opportunity.



Scalable by geography, crop and buyer segment.



Value created: better aggregation, lower friction, more traceability.

Scale levers: Orders ↑ GMV ↑ Repeat buyers ↑ Route utilization

CHALLENGES & MITIGATION



Digital Literacy

Local language +
simple UI Voice
interface



Connectivity

Offline /
Low-bandwidth
design



Payment Trust

Escrow • Delivery
Confirmation •
Automated Settlement



Logistics
Availability

Multiple verified
partners



Cold – Chain
Limitation

Suitable vehicle
+ priority routing

Impact and Benefits

"Better Prices. Less Waste. Greater Impact."

FARMER BENEFITS



**25-35%
Higher
income**

Better price no
middle man



**Market
Access**

Direct connect
to bulk buyers
and consumers



**Reduced
Distress**

Sell at right
time, right
place.



**Guaranteed
Payment**

Payment
released after
delivery.

REPORT



GitHub repository

SOCIAL IMPACTS



**Less Farmer
Distress**

Better price
reduce
dependency on
moneylenders.



**Women
Empowerment**

64% of
agri-workforce, FPO
model gives direct
income.



**Less Rural
Migration**

Stable from
income reduces
migration to
cities.

CONSUMER BENEFITS



**Lower
Prices**

No middleman
= affordable
produce.



**Fresher
Produce**

Direct farm
to door
delivery.



**Price
Stability**

AI forecasting
reduce price
shocks.



**Full
Traceability**

Digital record of
food origin.

ENVIRONMENTAL IMPACTS



**Less Food
Waste**

Efficient logistics
reduces 40% pre-
market spoilage.



**Lower
Emissions**

Route optimization
cuts fuel use &
carbon footprints.



**Less Methane
Release**

Less rotting food = less
decomposition based
methane release.

Research Papers & Industry Evidence

[1] Farmer's Share – RBI (2024) [\[Link\]](#)

Farmers' share in consumer price: Tomato 33%, Onion 36%, Potato 37%.

[2] Post-Harvest Losses – MoFPI / NABCONS (2022) [\[Link\]](#)

Fruit losses: 6.02–15.05%; Vegetable losses: 4.87–11.61%.

[3] Digital Agriculture – Agricultural Economics Research Review (2024) Digital platforms such as e-NAM/ONDC can improve market access, transparency & price realization. [\[Link\]](#)

[4] AI in Supply Chain – Research Paper [\[Link\]](#)

ML-based demand forecasting supports better demand prediction and supply-chain planning

Government Data

[5] e-NAM – Government of India (2026) [\[Link\]](#)

1,656 mandis integrated across 23 States + 4 UTs as of March 2026.

[6] Agriculture Census – Government of India [\[Link\]](#)

Official Agriculture Census source for operational holdings and size-class data.

[7] Rural Internet Connectivity – TRAI (2024) [\[Link\]](#)

398.35 million rural internet subscribers as of March 2024.

[8] AgriStack – Ministry of Agriculture & Farmers Welfare [\[Link\]](#)

Government digital infrastructure aimed at improving data access, digital services and farmers' market access.

Our Survey

[Click Here](#)

We conducted a survey on
Sept 2026

(50+ respondents – Farmers,
Traders, Consumers)



Our Prototype

[Click Here](#)

Competitors Comparison

Features	eNAM	DeHaat	BigHaat	KrishiSetu
Farmer–Buyer Connect	✓	✓	✗	✓
Integrated Logistics	✗	✗	✓	✓
Price Transparency	✗	✓	✓	✓
Multilingual Access	✗	✓	✗	✓
Escrow Payments	✗	✗	✗	✓
Real-time Market Data	✗	✓	✓	✓

Primary Research (Survey)

92% of respondents reported a lack of market price information



88% mention transport and logistics as a major challenges



95% prefer direct connection with verified buyers



76% Face quality and wastes issue during transport / storage

